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Concise Report

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Sequence [2,3]-Sigmatropic Rearrangement: One-Pot Synthesis of Propargyl Allenylamines

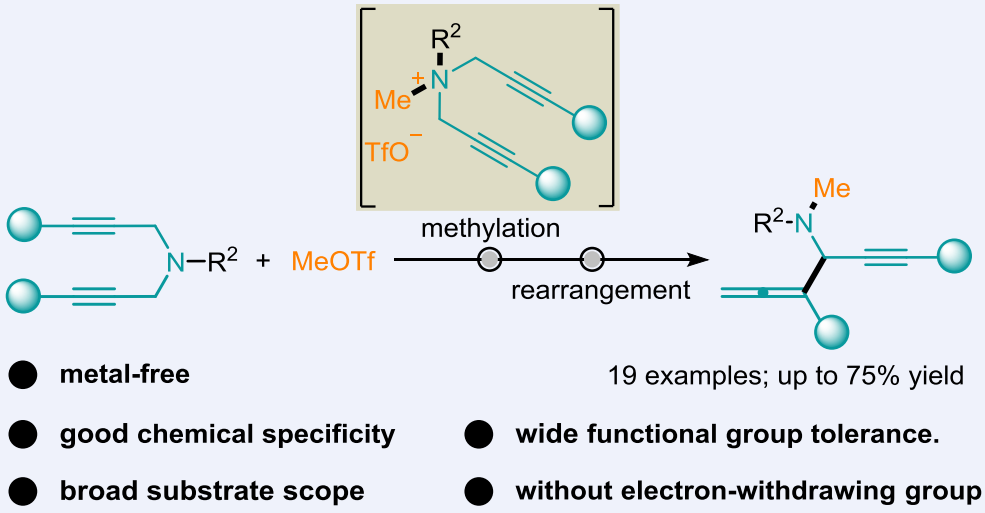
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Keywords

Allene | Propargylamine | C—N activation | Sigmatropic rearrangement | Metal-free | Synthetic methods | Domino reactions

Comprehensive Summary



Allenes, served as highly sought-after building blocks, are an indispensable component of synthetic chemistry. Their utility in modulating the chemical, physical, and pharmaceutical properties of organic compounds make allenes a desirable choice in various applications. Here, we report a facile method for the atom-economical synthesis of propargyl allenylamines via an underdeveloped [2,3]-sigmatropic rearrangement. Our strategy employs easily accessible propargylamines as starting materials, which are first converted into propargyl ammonium salts, followed by a base-promoted [2,3]-sigmatropic rearrangement. This one-pot, two-step reaction proceeds in the absence of transition metals, displays a very broad scope, and does not require the introduction of the electron-withdrawing group into the starting materials.

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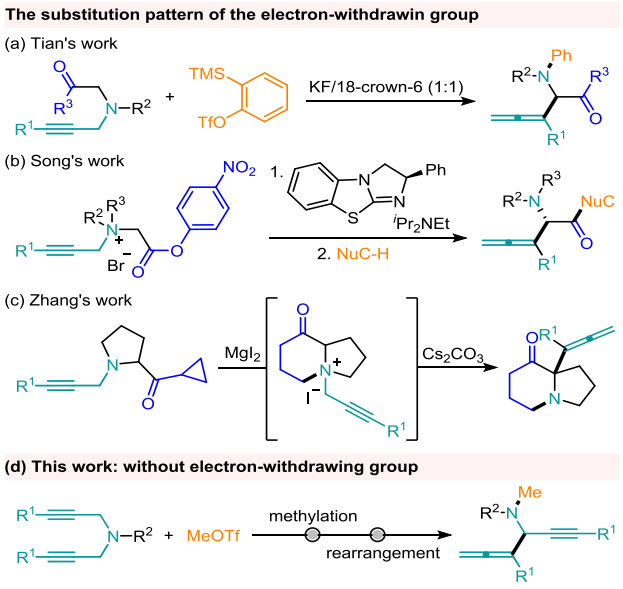
Background and Originality Content

Allenes and their derivatives, which are powerful and versatile building blocks in synthetic chemistry, have been widely used for the synthesis of various biologically active heterocycles and structurally-complex molecules.[1] These compounds also have a broad spectrum of biological activities and are found in a large number of natural products such as enprostil, mycomycin, acalycixeniolides E, archangiumide, and so on.[2] Consequently, a number of synthetic approaches have been established to construct allene-containing molecules.[3] Notably, transition metal-mediated allenation of terminal alkynes via subsequent 1,5-H transfer-β-elimination[4] and transition metal-catalyzed electrophilic substitutions of different propargylic electrophiles[5] represent the elegant strategies towards a diverse range of allene compounds.

In recent years, molecular rearrangements such as [2,3]-sigmatropic rearrangement and Claisen rearrangement have been successively developed to prepare functionalized allenes with high atom- and step-economy.[6] In this field, the [2,3]-sigmatropic rearrangement of propargylamines is an enduring challenge and has attracted considerable attention. In such processes, a specific type of allene, called allenylamine, is produced by the use of different reaction systems. For example, in 2018, Tian and co-workers reported a benzyne-mediated [2,3]-sigmatropic rearrangement of tertiary propargylamines to synthesize allenylamines using 2-(trimethylsilyl)aryl triflate as an aryne precursor (Scheme 1a).[7] In the same year, Song and co-workers disclosed a catalytic asymmetric [2,3]-sigmatropic rearrangement of propargyl ammonium salts in the presence of the chiral isothiourea catalyst (Scheme 1b).[8] More recently, a sequence of MgI_2-promoted ring opening of cyclopropyl ketones and base-promoted [2,3]-sigmatropic rearrangement of propargyl ammonium ylides has been developed by the Zhang group, yielding a variety of allenyl-substituted indolizidines (Scheme 1c).[9a] However, these methods rely on the introduction of electron-withdrawing groups at the α-position of the nitrogen atom and typically exploit the acidity of the C–H group next to an electron-withdrawing group functionality for domino deprotonation/[2,3]-sigmatropic rearrangement. It is noteworthy that the [2,3]-sigmatropic rearrangement, which only involves less acidic propargylamine derivatives and avoids the substitution pattern of the electron-withdrawing group, remains an uncharted territory.

Inspired by our recent study on the chemical transformations of propargylamines,[10] we envisaged that the synthesis of structurally diverse allenylamines could be achieved by molecular rearrangement of functionalized propargylamines. In this article, we disclose a sequential strategy of nucleophilic substitution of dipropargylamines with methyl trifluoromethanesulfonate (MeOTf) and [2,3]-sigmatropic rearrangement of quaternary propargylic ammonium ylides under mild conditions (Scheme 1d). This protocol correctly employs the appropriate dipropargylamines as starting materials, which can aid in the development of an efficient [2,3]-sigmatropic rearrangement without the introduction of the electron-withdrawing groups. The study represents the first example of the molecular rearrangement of dipropargylamine, leading to the generation of propargyl allenylamines in a one-pot manner. The synergistic effect of the allene and propargyl functional groups may allow this type of compound to be considered as a unique functionality with special reactivity.[11]

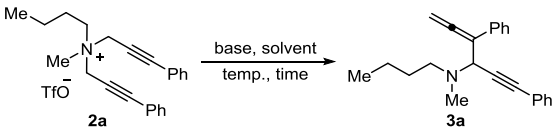
Scheme 1 [2,3]-Sigmatropic rearrangement of propargylaime derivative



Results and Discussion

To evaluate the feasibility of the reaction, we initially synthesized the model substrate, dipropargyl ammonium salt **2a**, through a copper-catalyzed A^3-coupling reaction to obtain dipropargylamine,[12] which was then followed by a nucleophilic substitution with MeOTf.[13] Subsequently, compound **2a** was introduced into the reaction system under the catalyst-free conditions, leading to the formation of the desired product **3a** in 17% yield in the presence of KOH (1.0 equiv) and tetrahydrofuran (THF) at 65 °C for 6 h (Table 1, entry 1). Buoyed by this initial success, a series of solvents were tested to optimize the yield of product **3a** (entries 2—4). The results revealed that toluene was the most effective solvent, delivering a 22% yield of the target product (entry 4). Next, various bases including NaOH, Na_2CO_3, and *t*-BuOK were screened (entries 5—7). *t*-BuOK was identified to be optimal, providing **3a** in 54% yield (entry 7). To our delight, the yield of the product **3a** was further improved to 75% when the amount of *t*-BuOK was reduced to 0.8 equiv (entries 8 and 9). However, further optimization of reaction parameters was not effective in promoting the reaction (entries 10—13). For example, modulating the reaction temperature resulted in the generation of desired

Table 1 Optimization of [2,3]-sigmatropic rearrangement[a]



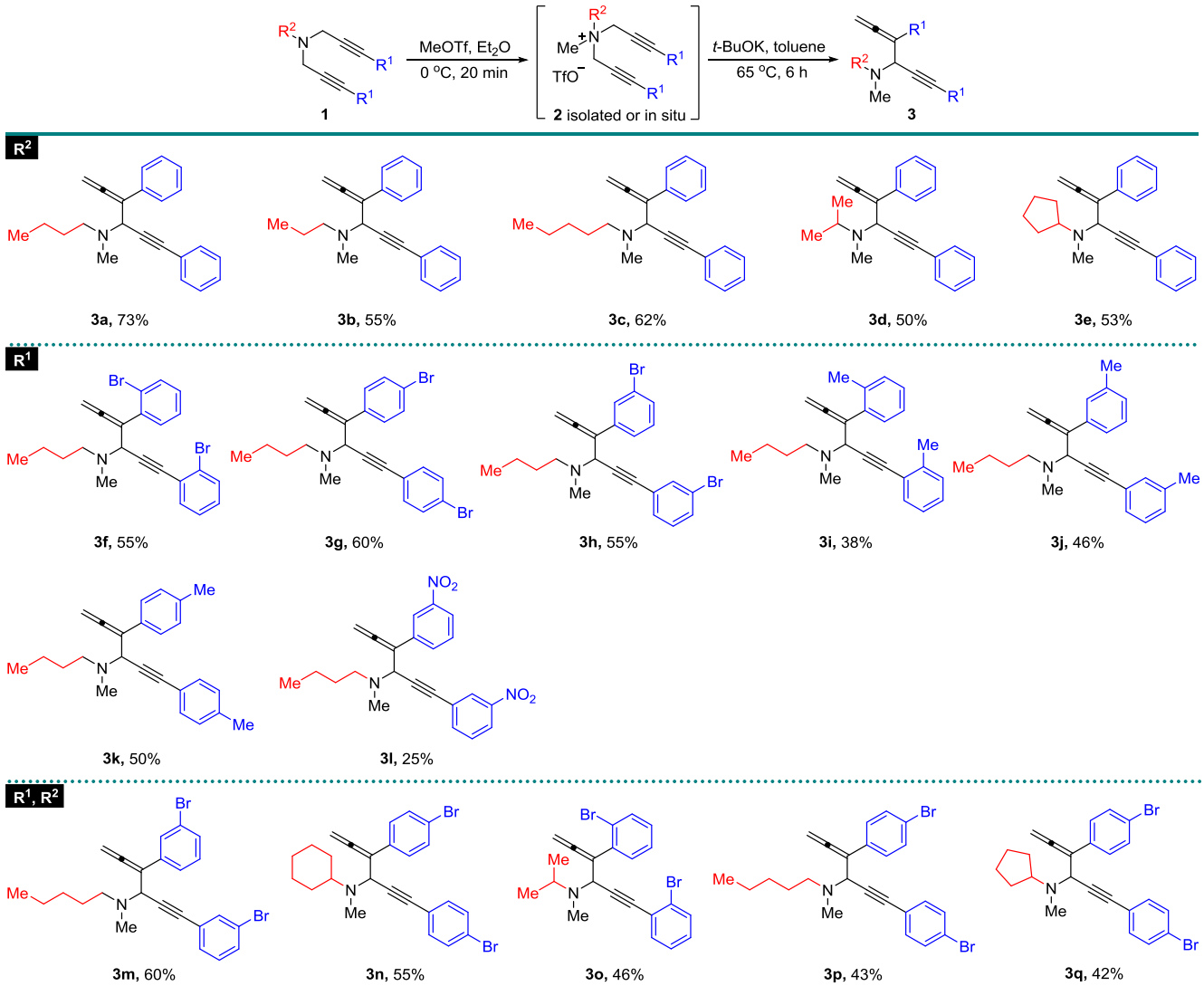
Entry	Base/mmol	Solvent	Temp/°C	Time/h	Yield[b]/%
1	KOH (1.0)	THF	65	6	17
2	KOH (1.0)	1,4-dioxane	65	6	14
3	KOH (1.0)	chlorobenzene	65	6	19
4	KOH (1.0)	toluene	65	6	22
5	NaOH (1.0)	toluene	65	6	21
6	Na_2CO_3 (1.0)	toluene	65	6	16
7	*t*-BuOK (1.0)	toluene	65	6	54
8	*t*-BuOK (0.8)	toluene	65	6	75
9	*t*-BuOK (0.6)	toluene	65	6	35
10	*t*-BuOK (0.8)	toluene	80	6	65
11	*t*-BuOK (0.8)	toluene	rt	6	55
12	*t*-BuOK (0.8)	toluene	65	4	23
13	*t*-BuOK (0.8)	toluene	65	20	40
14[c]	*t*-BuOK (0.8)	toluene	65	6	51

[a] Reactions conditions: **2a** (0.5 mmol), base (0.6—1.0 mmol) stirred in solvent (2 mL) for 4—20 h at rt—80 °C. [b] Isolated yield. [c] In a nitrogen atmosphere.

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Scheme 2 Scope of propargylamines[a]



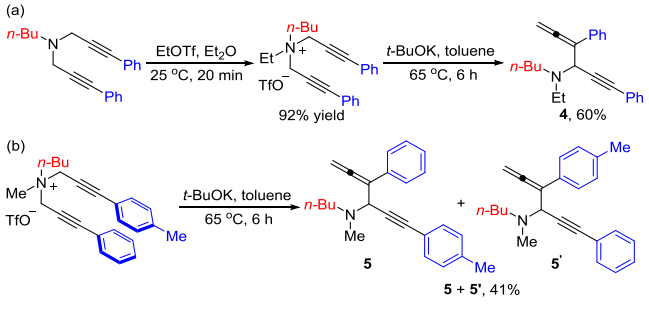
[a] Reaction conditions: propargylamines **1** (0.5 mmol), MeOTf (1.0 mmol) stirred in Et_2O (4 mL) for 20 min at 0 °C, then removing Et_2O and adding *t*-BuOK (0.8 mmol), toluene (2 mL), the mixture stirred for 6 h at 65 °C; Isolated yield.

product **3a** in 65% and 55% yields, respectively (entries 10 and 11). Changing the reaction time led to a significant decrease in the yield to 23% and 40%, respectively (entries 12 and 13). Furthermore, a reduction in the yield of **3a** was observed when the reaction was conducted in a nitrogen atmosphere (entry 14).

Having established the optimal reaction parameters, we explored its versatility in the [2,3]-sigmatropic rearrangement using a diverse array of propargylamines **1** via a one-pot, two-step procedure. As depicted in Scheme 2, our approach exhibited a robust tolerance for various functional groups and generated the desired products **3a—3q** in moderate to good yields. For example, propargylamines bearing different R^2 groups, including straight-chain alkyl (propyl, butyl, amyl), branched alkyl (isopropyl), and cycloalkyl (cyclopentyl), proceeded smoothly to deliver the corresponding products **3a—3e** in good yields. Furthermore, a variety of R^1-substituted propargylamines were screened and the reaction worked efficiently to give the desired products **3f—3l** in moderate to good yields. In these cases, both electron-withdrawing and electron-donating functional groups, including the synthetically useful bromo substituent, were accommodated in the different positions of the benzene ring, while the nitro-subsitituted substrate led to the formation of **3l** in a slightly reduced yield. To further investigate the functional group tolerance of the method, other R^1- and R^2-substituted propargylamines were introduced into the reaction. The results showed that these substrates were compatible with the reaction, allowing the synthesis of the target products **3m—3q** in good yields.

In addition, when the ethyl trifluoromethanesulfonate (EtOTf) was chosen as the alkylation reagent, the desired [2,3]-sigmatropic rearrangement product **4** was obtained in 60% yield (Scheme 3a). Unsymmetrical dipropargyl ammonium salt was also applicable, however, leading to the formation of a mixture of two regioisomers **5** and **5'** in 42% yield (Scheme 3b).

Scheme 3 Scope of dipropargyl ammonium salts

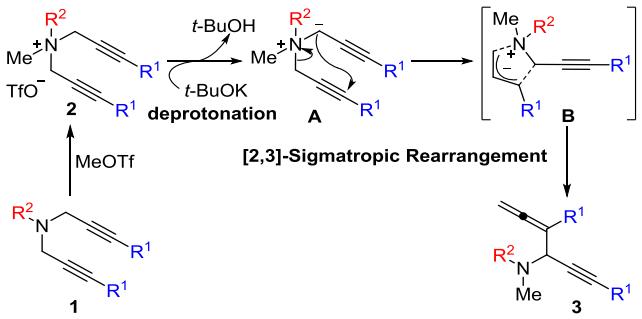


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Based on our experimental observations and the previous literature,[7,9] we proposed a plausible reaction pathway (Scheme 4). At first, the reaction began with the generation of dipropargyl ammonium salt **2** through a nucleophilic substitution of dipropargylamine **1** with methyl trifluoromethanesulfonate. Then, the intermediate **2** underwent deprotonation to form quaternary dipropargylic ammonium ylide **A** in the presence of *t*-BuOK. Finally, the [2,3]-sigmatropic rearrangement of ylide **A** was achieved through a converted five-membered transition state **B**, resulting in the formation of the desired product **3**.

Scheme 4 Plausible reaction pathway



Conclusions

In summary, we have established a unique [2,3]-sigmatropic rearrangement of dipropargylamines, which is currently underdeveloped, offering a straightforward and modular method for the synthesis of a wide range of propargyl allenylamines. Our research reveals an atom-economical and versatile reaction with a broad scope of substrates without the introduction of the electron-withdrawing groups, outstanding efficiency, and high chemoselectivity. The distinctive combination of propargyl and allenyl groups holds substantial potential as a robust framework for the construction of valuable compounds.

Supporting Information

The supporting information for this article is available on the WWW under https://doi.org/10.1002/cjoc.202400307.

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